

Acids and Bases

Acids	Bases
Strong Acids Hydrochloric acid, HCl Hydrobromic acid, HBr Hydroiodic acid, HI Nitric acid, HNO ₃ Sulfuric acid, H ₂ SO ₄ Perchloric acid, HClO ₄	Strong Bases <i>Group 1 hydroxides:</i> Lithium hydroxide, LiOH Sodium hydroxide, NaOH Potassium hydroxide, KOH Rubidium hydroxide, RbOH Cesium hydroxide, CsOH <i>Heavy Group 2 hydroxides:</i> Calcium hydroxide, Ca(OH) ₂ Strontium hydroxide, Sr(OH) ₂ Barium hydroxide, Ba(OH) ₂
Weak Acids (a few of many examples) Hydrofluoric acid, HF Phosphoric acid, H ₃ PO ₄ Acetic acid, HC ₂ H ₃ O ₂ (or CH ₃ COOH)	Weak Bases (one of many examples) Ammonia, NH ₃

Rules for naming acids:

- **with a monatomic anion (e.g. Cl⁻ in HCl):** *hydro-* + [ROOT] + *-ic acid* (e.g. hydrochloric acid)
- **with a polyatomic anion (e.g. NO₃⁻ in HNO₃):**
ROOT-ate becomes *ROOT-ic acid* — e.g. the nitrate anion becomes nitric acid, sulfate becomes sulfuric acid
ROOT-ite becomes *ROOT-ous acid* — e.g. the nitrite anion becomes nitrous acid, sulfite becomes sulfurous acid